

Multiple Choice

1. **Answer:** D

Options: A) 2:3 periodic ratio of Neptune:Pluto; B) Kirkwood Gaps.; C) Gaps in Saturn's rings.; D) Breaking of small Jovian moons to form ring materials.

Note: Correct option: D - Breaking of small Jovian moons to form ring materials.

2. **Answer:** D

Options: A) The corona is made up mostly of charged particles which aren't luminous.; B) It's much too cool to emit visible light; C) We can't see magnetic fields; D) It's too diffuse

Note: Correct option: D - It's too diffuse

3. **Answer:** B

Options: A) pieces of comets rather than of asteroids.; B) chunks of large differentiated asteroids that were shattered by collisions.; C) chunks of rock chipped off the planet Mars.; D) leftover chunks of rock from the earliest period in the formation of the solar system.

Note: Correct option: B - chunks of large differentiated asteroids that were shattered by collisions.

4. **Answer:** C

Options: A) a New Horizons flyby in the 1990 s; B) Hubble Space Telescope images that resolved Pluto's disk; C) brightness measurements made during mutual eclipses of Pluto and Charon; D) radar observations made by the Arecibo telescope

Note: Correct option: C - brightness measurements made during mutual eclipses of Pluto and Charon

5. **Answer:** C

Options: A) It has a fusion crust; B) It contains solidified spherical droplets; C) It is highly processed; D) It has different elemental composition than earth

Note: Correct option: C - It is highly processed

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Cassiopeia'.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Power emitted is 1/16 times as high; peak emission wavelength is 2 times longer.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** Leo. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key